Department of Physics

53080 - Advanced Solid State Physics

Term Project:

Ferroelectric Materials and Ferroelectric Random Access Memory (FeRAM) [1]

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December 4, 2025

[1] **Declaration of AI aids and -tools:** Text: *Grammarly* was used for spell checking, and *ChatGPT* helped with rephrasing and additional spell checking.

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1 Introduction

This report is a term project in the course Advanced Solid State Physics at Chung-Ang University. I chose the topic of ferroelectric materials and ferroelectric random access memory (FeRAM) because, in an earlier integrated circuits course at NTNU, I worked with circuit-level simulations of memory cells using AIM-Spice and Verilog. FeRAM provides a natural link between that device-oriented view and the solid state physics covered in this course.

In simple terms, FeRAM is a type of non-volatile memory where information is stored using the electric polarization of a ferroelectric material instead of charge on a floating gate. This makes it an interesting alternative to Flash and EEPROM in embedded systems, especially when data has to be updated frequently [1].

The aim of this report is to review the ferroelectric physics that is relevant for memory applications and to explain how these concepts are realised in FeRAM at the device, circuit and chip level. As a concrete example, an embedded FeRAM microcontroller from the MSP430FR59xx family by Texas Instruments is briefly discussed. [2] The text is organised as follows. Section 2 provides the theoretical background on ferroelectricity and hysteresis. Section 3 introduces typical ferroelectric materials used in memories and their main requirements. Section 4 describes the operation of the 1T–1C FeRAM cell and its organisation into memory arrays, and Section 5 considers chip-level integration and an embedded FeRAM microcontroller. Section 6 gives a short discussion and outlook, and Section 7 concludes the report.

2 Theoretical Background

Ferroelectric materials form a special class of dielectric materials. In a normal dielectric, an applied electric field induces an electric polarization, which disappears when the field is removed. In a ferroelectric material, there is a spontaneous polarization even without an external field, and this polarization can be switched between at least two stable orientations by an applied field [3].

In this section, the key ideas needed to understand ferroelectric memories are reviewed. The focus is on quantities that later appear in device and circuit descriptions, such as polarization, electric field, coercive field and hysteresis.

2.1 Ferroelectricity in Dielectrics

The macroscopic electric polarization P of a solid is defined as the dipole moment per unit volume. In an ordinary linear dielectric, the polarization is proportional to the applied electric field E,

$$P = \varepsilon_0 \chi E, \tag{1}$$

where ε_0 is the vacuum permittivity and χ is the electric susceptibility. The relative dielectric constant is $\varepsilon_r = 1 + \chi$ [3, p. 459].

Ferroelectric materials are also dielectrics, but below a critical temperature, the Curie temperature, they exhibit a spontaneous polarization P_s even when $E = 0$. The crystal is divided into domains with different polarization directions, and the macroscopic

polarization depends on how these domains are distributed. Above the Curie temperature the material is paraelectric with higher symmetry and $P_s = 0$, while below it a lower-symmetry ferroelectric phase with $P_s \neq 0$ appears. This behaviour can be described phenomenologically by a Landau-type free energy with two equivalent minima below the Curie temperature. [3]

2.2 Polarization–Electric Field Hysteresis

A characteristic feature of ferroelectrics is that the polarization–electric field (P–E) relation shows hysteresis. If the electric field is cycled between positive and negative values, the polarization follows a closed hysteresis loop. At large positive fields the polarization saturates at $+P_s$, and when the field is reduced to zero a finite remanent polarization P_r remains. A negative field of magnitude E_c (the coercive field) is needed to switch the polarization to $-P_s$. The process is repeated in the opposite direction when the field is increased back to positive values.

The important point is that at zero applied field, $E = 0$, the material can be in either of two stable states with polarization $+P_r$ or $-P_r$, depending on the previous maximum field. This bistability and hysteresis mean that the material "remembers" the history of the applied field even after the field is removed. In ferroelectric memories, these two stable remanent polarization states are then associated with the logical states "0" and "1".

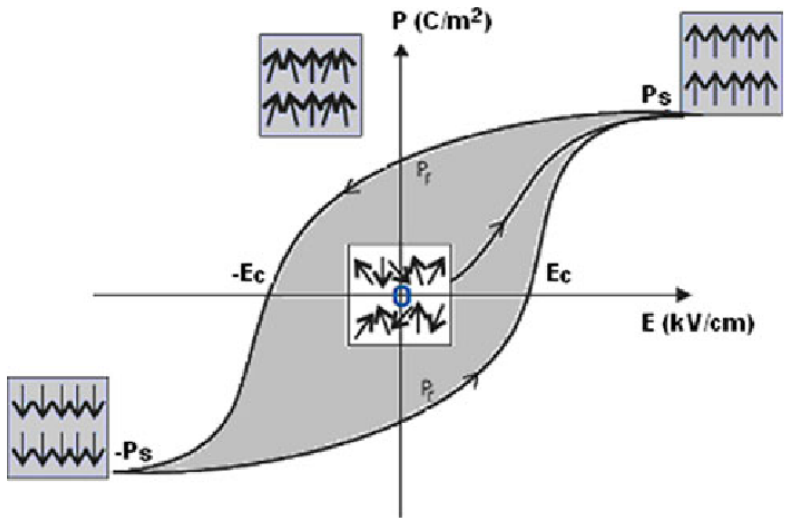

Figure 1: Schematic hysteresis loop for a ferroelectric material, indicating the remanent polarization P_r and coercive field E_c. Figure adapted from [4].

Figure 1 illustrates the saturation, remanent polarization and coercive field in a ferroelectric material. The two intersection points with the vertical axis ($E = 0$) correspond to the stable remanent polarization states that can be used to represent binary information in a ferroelectric memory cell.

3 Ferroelectric Materials for Memory Applications

The practical use of ferroelectricity in non-volatile memories depends not only on the general phenomena described in Section 2, but also on the properties of specific materials. In ferroelectric random access memory (FeRAM), thin-film or ceramic ferroelectrics must combine switchable polarization with acceptable dielectric loss, controlled conduction and compatibility with standard processing. [3] Recent work has focused in particular on lead-free compounds that still offer sufficient remanent polarization and low loss for device applications. [5, 6]

3.1 Typical Ferroelectric Materials

Many ferroelectric materials belong to the perovskite or related oxide families. A classic textbook example is barium titanate ($BaTiO_3$), where the ferroelectric phase appears below the Curie temperature when the Ti ion moves off-centre inside the oxygen octahedron, leading to a spontaneous polarization along a crystal axis. [3]

A second important class are Aurivillius oxides such as $Bi_4Ti_3O_{12}$ (BIT), which consist of alternating $(Bi_2O_2)^{2+}$ layers and perovskite-like slabs. These materials show large spontaneous polarization and have long been considered for non-volatile memory applications, but suffer from issues such as leakage and domain pinning. [5] Panda *et al.* recently studied selenium-doped bismuth titanate $Bi_4Ti_{2.9}Se_{0.1}O_{12}$, synthesized by a solid-state route, and found an orthorhombic Aurivillius structure with comparatively low dielectric loss at 1 MHz and a room-temperature ferroelectric hysteresis loop whose remanent polarization makes the compound suitable for memory applications. [5]

Another example of current interest is the brownmillerite compound $NaBiFe_2O_5$, a lead-free ceramic with triclinic crystal symmetry. [6] report a high dielectric constant, low loss tangent and a room-temperature P–E loop indicating ferroelectric behaviour, suggesting that $NaBiFe_2O_5$ is a promising candidate for electronic and energy-storage devices. [6]

3.2 Material Requirements for FeRAM

For use in FeRAM or related non-volatile memories, a ferroelectric thin film or ceramic must satisfy several requirements beyond simply having a non-zero spontaneous polarization. Important properties include:

- **Remanent polarization:** P_r must be large enough to generate a clear charge or polarization contrast between the two logic states. In $Bi_4Ti_{2.9}Se_{0.1}O_{12}$, the observed room-temperature hysteresis loop demonstrates a remanent polarization sufficient for memory operation, while maintaining low loss at high frequency. [5]
- **Coercive field and switching behaviour:** The coercive field E_c should be low enough to allow switching at voltages compatible with the circuit, but high enough to avoid unintended switching due to noise. The shape and saturation of the P–E loop, as reported for both $Bi_4Ti_{2.9}Se_{0.1}O_{12}$ and $NaBiFe_2O_5$, give insight into how easily domains can be reversed. [5, 6]
- **Dielectric loss and leakage:** Low dielectric loss and controlled conduction are essential for long retention times and low power consumption. Both $Bi_4Ti_{2.9}Se_{0.1}O_{12}$

and $NaBiFe_2O_5$ show relatively low loss over 1 kHz–1 MHz and temperature-dependent behaviour characteristic of thermally activated transport, which is important for reliable device operation. [5, 6]

These examples illustrate how structural details (Aurivillius vs. brownmillerite), dopants and processing conditions directly influence the dielectric and ferroelectric parameters that are critical for memory applications.

4 FeRAM Device Operation and Circuit Architecture

In a ferroelectric random access memory (FeRAM), information is stored in the polarization state of a ferroelectric capacitor or ferroelectric-gate transistor. From a circuit point of view, two main architectures are used: capacitor-type FeRAM cells, often with a 1T–1C structure, and FET-type FeRAM cells where a ferroelectric film is integrated directly in the gate stack of a MOSFET or in a 1T2C configuration. [1]

4.1 FeRAM Cell Structure

In capacitor-type FeRAM, the most common cell is the 1T–1C cell, consisting of one access transistor (1T) and one ferroelectric capacitor (1C). [1] The ferroelectric capacitor is connected between a *plate line* and a *bit line*, while the other terminal of the transistor is connected to the bit line. The gate of the access transistor is driven by a *word line*. When the word line is activated, the transistor turns on and couples the ferroelectric capacitor to the bit line.

The logical states "0" and "1" correspond to the two possible polarization directions of the ferroelectric capacitor at zero applied voltage. In practice, the state is sensed through the charge that flows during a read operation rather than by directly measuring polarization.

4.2 Write Operation

A write operation in a 1T–1C FeRAM cell is performed by applying a voltage pulse across the ferroelectric capacitor while the access transistor is on. To write a logical "1", the plate and bit lines are driven so that the electric field across the capacitor switches the polarization into one direction; to write a "0", the field polarity is reversed. [1] The write voltage must exceed the coercive field of the ferroelectric material but should not be much larger to avoid unnecessary stress and fatigue.

4.3 Read Operation

In capacitor-type FeRAM, the read operation is based on sensing the charge associated with polarization switching and is therefore *destructive*. A simplified sequence is as follows. [1] First, the bit line is precharged to a reference voltage. Next, the word line is activated so that the access transistor turns on, and a voltage pulse is applied to the plate line. If the stored polarization state is aligned with the applied field, only a small displacement charge flows and the bit line voltage changes little. If the polarization is opposite, it will switch during the pulse, leading to a larger transferred charge and a larger voltage change.

A sense amplifier detects this difference and decides whether the stored state corresponds to “0” or “1”.

Because reading involves switching the polarization in one of the cases, the original state of the capacitor is not preserved and must be written back immediately after the sense operation. This internal restore is part of the FeRAM timing.

In contrast, FET-type FeRAM cells, where a ferroelectric thin film is used directly in the gate stack, can in principle offer non-destructive read-out, at the cost of more demanding interface engineering. [1]

4.4 From Cell to Memory Array

In a complete memory array, many 1T–1C cells are arranged in rows and columns. Each row is selected by a word line that controls the access transistors of all cells in that row, while each column is connected to a bit line that carries the read and write signals. The other terminals of the ferroelectric capacitors are connected to plate lines, which can be driven per row, per block of rows or globally, depending on the architecture. [1]

4.5 Key Performance Characteristics

Important figures of merit for a FeRAM device include access time, endurance, retention, operating voltage and energy per operation. In FeRAM arrays based on bismuth-layered perovskites, Ishiwara reports fast operation and stable on/off ratios in FET-type cells with $SrBi_2Ta_2O_9$ gate stacks, illustrating that ferroelectric switching can be exploited at practical voltages. [1] Combined with the material parameters discussed in Section 3, these circuit-level characteristics determine which applications are best suited for FeRAM.

5 Chip-Level Implementation and Example Application

In a complete integrated circuit, FeRAM is combined with logic, analogue blocks and other memories to form a system-on-chip. This section briefly outlines FeRAM integration in CMOS technology and illustrates its use with an embedded FeRAM microcontroller.

5.1 Integration with CMOS Technology

Integrating a ferroelectric layer into a CMOS process requires additional processing steps beyond the standard transistor flow. In capacitor-type FeRAM, this typically means forming the access transistors with conventional CMOS processing and then depositing and patterning the ferroelectric thin film and its electrodes in the back-end-of-line (BEOL). In FET-type FeRAM, a ferroelectric film such as $SrBi_2Ta_2O_9$ (SBT) is integrated directly into the gate stack on Si, often with a buffer layer to control the interface. [1]

The ferroelectric film must be deposited with controlled composition, thickness and crystal structure. Techniques such as sol–gel and related chemical routes are widely used for bismuth-layered and perovskite-type ferroelectrics, since they allow good control over phase formation after suitable heat treatments. [4] At the same time, the choice of electrode materials must ensure good electrical contact and stable interfaces while remaining

compatible with aluminium or copper interconnects and the overall thermal budget of the process.

A key constraint in integration is that the annealing steps needed to obtain the desired ferroelectric phase and domain structure must not degrade the underlying CMOS devices or interconnects. In SBT-based FET-type FeRAM, for example, the use of buffer layers and carefully optimised annealing conditions has been shown to yield stable on/off ratios over long endurance testing. [1] Despite these constraints, qualified processes for embedded FeRAM exist, so from a circuit designer's point of view the FeRAM block largely resembles embedded Flash or EEPROM, but with different timing and energy behaviour.

5.2 Embedded FeRAM in Microcontrollers

Embedded FeRAM is particularly attractive in low-power and battery-powered microcontrollers. Compared to embedded Flash, FeRAM can offer faster writes, lower write energy and higher endurance, which is beneficial when non-volatile data is updated frequently. In such devices, the CPU core connects via an internal bus to FeRAM, SRAM, peripherals and I/O pins. From the programmer's perspective, FeRAM is accessed through memory addresses in the same way as Flash, but write operations can be performed with much lower energy and without long erase cycles, making it easier to log sensor data and update parameters without strongly impacting battery life. [2]

5.3 Case Study: MSP430FR59xx FeRAM Microcontroller

As a concrete example, the MSP430FR59xx and MSP430FR58xx microcontroller family from Texas Instruments includes a 16-bit ultra-low-power CPU core, FeRAM, SRAM and mixed-signal peripherals. [2] A representative device such as the MSP430FR5969 provides up to 64 kB FeRAM and 2 kB SRAM, operates up to 16 MHz and runs from about 1.8–3.6 V. [2]

The FeRAM is used for both code and data and can be written at full clock speed without dedicated program or erase cycles. According to the datasheet, it offers very high endurance and significantly lower write energy than typical embedded Flash, which directly reduces energy consumption for frequent non-volatile writes. [2] The MSP430FR59xx devices target applications such as energy meters, industrial sensors and battery-powered or energy-harvesting systems, where the combination of ultra-low-power operation and FeRAM-based non-volatile memory enables robust data logging and configuration storage.

6 Discussion

Compared to charge-based Flash memories, FeRAM and related FeRAM technologies use polarization switching in ferroelectric oxides, which enables fast, symmetric read/write access and high endurance. [1] Recent work on lead-free Aurivillius and brownmillerite ceramics, such as $Bi_4Ti_{2.9}Se_{0.1}O_{12}$ and $NaBiFe_2O_5$, illustrates how doping and microstructure can be used to tune dielectric loss and remanent polarization for device applications. [5,6] At the same time, the practical MSP430FR59xx family shows that embedded FeRAM can already be used in low-power microcontrollers. [2]

7 Conclusion

Ferroelectric materials provide a natural physical basis for non-volatile memory, because their two stable polarization states can be used to store binary information. In this report, I have reviewed how these properties appear in real materials and how they can be exploited in FeRAM devices, from the level of ferroelectric thin films to the 1T–1C memory cell and its organisation into arrays. I also briefly discussed how FeRAM can be integrated in a microcontroller and used as embedded non-volatile memory in low-power systems. Overall, FeRAM appears particularly well suited for embedded and mixed-signal applications where frequent non-volatile updates, endurance and energy efficiency are more important than achieving the highest possible storage density.

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